

Two-dimensional Line Spectral Estimation for Constant Modulus Signals via Convex Optimization

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Abstract—Two-dimensional line spectral estimation is a fundamental problem in signal processing, with wide-ranging applications in radar systems and wireless communications. A prominent class of convex methods is atomic norm approaches, where the decoupled atomic norm method (DANM) offers good tradeoffs between computational speed and accuracy in the single-channel setting. In many practical applications, the underlying spectral-sparse signals also possess the constant modulus (CM) property, and multichannel measurements are available. However, since DANM is developed for the single-channel case, it is not tailored to exploit this additional structure to yield further performance gains. We therefore study how to effectively exploit the CM prior for multichannel frequency recovery from partial, regularly spaced time samples within a convex framework. To address this problem, we propose a modified version of DANM, termed DANM-CM. We demonstrate analytically that under a mild condition, the new algorithm successfully recovers the true frequency components. Extensive numerical experiments confirm the effectiveness of the proposed method.

Index Terms—Line spectral estimation, constant modulus, decoupled atomic norm minimization, multichannel sparse recovery.

I. INTRODUCTION

In the context of statistical signal processing, line spectrum estimation (LSE) addresses the problem of retrieving sparse spectral components from discrete time samples. Multidimensional LSE has attracted considerable attention in recent years, with two-dimensional (2D) formulations being particularly relevant to radar systems and wireless communications [1]–[3]. When multiple observations arise in such scenarios and share the same sparse frequency components, exploiting this joint sparsity improves recovery performance—a framework known as multichannel sparse recovery. Furthermore, many synthetic signals—particularly most digitally modulated signals—exhibit the constant modulus (CM) property in addition to sparsity [4], [5]. Previous works in array processing [6]–[8] have shown that incorporating CM prior yields significant performance gains. Although several techniques [9]–[12] have been developed to exploit CM structure in one-dimensional settings, the multi-dimensional case remains understudied. Motivated by this gap, this paper focuses on 2D multichannel spectral-sparse recovery that incorporates the CM property.

Research on the CM property dates back to the 1980s and has gained renewed momentum in recent years. Recent efforts focus on devising methods that exploit CM under different problem settings. Existing studies can be broadly divided into two categories: compressed sensing (CS)–based approaches and non-CS approaches. The constant modulus algorithm and its variants [13], [14] are stochastic, gradient-based methods that iteratively minimize a cost function containing a modulus constraint, thereby driving the outputs toward constant amplitude; they often require careful step-size tuning and may converge slowly or to undesirable stationary points. In contrast, the analytical constant modulus algorithm (ACMA) explicitly uses the model order and exploits the algebraic structure of the CM property by converting it into an eigenvalue problem, whose solution yields a separation matrix enforcing constant modulus on the recovered signals [12]. On the grid-based CS side, $\ell_{\infty,1}$ minimization replaces the ℓ_2 norm of each row (corresponding to one grid frequency across channels) with the ℓ_{∞} norm, simultaneously enforcing joint sparsity and aligning with the CM structure [11]. Within the gridless CS, the structured matrix recovery technique (SMART) provides an alternative characterization of the CM property and reformulates LSE problem as a non-convex structured matrix recovery problem with rank constraints [9]. The structured low-rank approximation-based atomic norm minimization for constant amplitude frequency estimation (SACA) can be regarded as a convex relaxation of SMART [10]. Within the atomic norm method (ANM) framework, it designs an atomic set with an explicit CM constraint and eliminates the need for prior model-order information. Both SMART and SACA typically involve large structured Hankel-Toeplitz block matrices, which can lead to large-scale matrix operations in iterative solvers.

Driven by demands from industrial applications, many one-dimensional techniques have been extended to the two-dimensional setting. Subspace methods offer high resolution and have been extensively studied; however, they typically rely on accurate model-order selection and can be sensitive to finite-snapshot effects and source coherence [15], [16]. The grid-based CS methods readily accommodate nonuniform sampling and scenarios with few channels, and they dispense with prior knowledge of the model order; however, they suffer from off-grid mismatch and their complexity grows rapidly with the grid density [17], [18]. Among gridless CS methods,

2D ANM is representative: it likewise avoids model-order tuning and enables super-resolution, but at significant computational expense [19]. Decoupled ANM (DANM) mitigates this burden by separating the 2D frequency information into two separate one-level Toeplitz matrices, thereby retaining the benefits of ANM while markedly reducing complexity [20]. Nevertheless, DANM is inherently tailored to single-channel observations, which limits its ability to fully leverage the CM property.

In this paper, we develop a DANM-based approach for the 2D LSE problem with the CM property, termed DANM-CM. We establish a theoretical guarantee that, under a frequency separation condition, DANM-CM recover exactly the true frequencies. Numerical experiments are conducted to validate the effectiveness of the proposed approach.

II. PROBLEM FORMULATION AND PRIOR ART

A. Problem Formulation

Without loss of generality, the model of 2D LSE can be expressed as: $\mathbf{X}_\kappa = \sum_{k=1}^K s_k e^{i2\pi \mathbf{h}_k^T \kappa}$, $\kappa \in \mathcal{K}$, where $\mathbf{h}_k = [f_k, \phi_k] \in [0, 1]^2$ denotes the 2D frequency pair, $s_k \in \mathbb{C}$ is amplitude of the k th spectral component, and $\mathcal{K} \in [N] \times [M]$ is the 2D integer grid of time indices. The objective is to recover frequency components of K signals from partial time-domain samples. The data matrix can be rewritten as: $\mathbf{X} = \sum_{k=1}^K s_k \mathbf{a}_N(f_k) \mathbf{a}_M(\phi_k)^T = \mathbf{A}_N(\mathbf{f}) \text{diag}(\mathbf{s}) \mathbf{A}_M(\boldsymbol{\phi})^T \in \mathbb{C}^{N \times M}$, where $\mathbf{A}_N(\mathbf{f}) = [\mathbf{a}_N(f_1), \dots, \mathbf{a}_N(f_K)]$, $\mathbf{a}_N(f_k) = [1, e^{i2\pi f_k}, \dots, e^{i2\pi f_k(N-1)}]^T$. Furthermore, we consider the multichannel setting in this paper. The multichannel model in matrix form is given by:

$$\begin{aligned} \mathbf{X}(l) &= \sum_{k=1}^K s_k(l) \mathbf{a}_N(f_k) \mathbf{a}_M(\phi_k)^T \\ &= \mathbf{A}_N(\mathbf{f}) \text{diag}(\mathbf{s}(l)) \mathbf{A}_M(\boldsymbol{\phi})^T \in \mathbb{C}^{N \times M}, l \in [L], \end{aligned} \quad (1)$$

where L is the channel number. Motivated by practical scenarios, we further consider the CM property, i.e., $|s_k(l)| = b_k$ for all $k \in [K]$ and $l \in [L]$. In addition, we assume that the observed data matrices are $\mathbf{X}_\Omega(l) \in \mathbb{C}^{n \times m}$, $l \in [L]$, corresponding to the entries indexed by $\Omega \subset [N] \times [M]$.

B. Decoupled Atomic Norm Minimization

This subsection reviews the DANM approach for 2D LSE. The ANM-based methods provide a general framework for obtaining signal sparse representations. It is natural to define the atomic set as $\mathcal{A} := \{\mathbf{a}_N(f) \mathbf{a}_M(\phi)^T : f \in [0, 1], \phi \in [0, 1]\}$. The atomic norm of $\mathbf{X}$ is defined as the gauge function associated with the convex hull of $\mathcal{A}$:

$$\begin{aligned} \|\mathbf{X}\|_{\mathcal{A}} &:= \inf\{v > 0 : \mathbf{X} \in v \text{conv}(\mathcal{A})\} \\ &= \inf\left\{\sum_k |s_k| : \mathbf{X} = \sum_k s_k \mathbf{a}_N(f_k) \mathbf{a}_M(\phi_k)^T, \right. \\ &\quad \left. \mathbf{a}_N(f_k) \mathbf{a}_M(\phi_k)^T \in \mathcal{A}, s_k \in \mathbb{C}\right\}. \end{aligned} \quad (2)$$

According to the [20, Theorem 4.2], under the separation condition that, in at least one frequency dimension (either f or ϕ), the wrap-around distance between any two distinct

frequencies is no smaller than $\frac{1.19}{\lfloor (N-1)/4 \rfloor}$ or $\frac{1.19}{\lfloor (M-1)/4 \rfloor}$, (2) is equivalent to the following semidefinite programming (SDP):

$$\begin{aligned} \|\mathbf{X}\|_{\mathcal{A}} &= \min_{\mathbf{t}_x, \mathbf{t}_y} \frac{1}{2N} \text{tr}(\mathcal{T} \mathbf{t}_x) + \frac{1}{2M} \text{tr}(\mathcal{T} \mathbf{t}_y) \\ \text{subject to } &\begin{bmatrix} \mathcal{T} \mathbf{t}_y & \mathbf{X}^H \\ \mathbf{X} & \mathcal{T} \mathbf{t}_x \end{bmatrix} \succeq 0. \end{aligned} \quad (3)$$

A key benefit of DANM is the reduced SDP size and the resulting computational savings. In particular, DANM decouples the 2D frequency information into two one-level Toeplitz matrices, yielding an $(N+M) \times (N+M)$ positive semidefinite (PSD) constraint. For comparison, a commonly used SDP formulation for vectorization-based 2D ANM [19] involves a block-PSD constraint of the form $\begin{bmatrix} \mathbf{S}(T) & \mathbf{x} \\ \mathbf{x}^H & t \end{bmatrix} \succeq 0$, where $\mathbf{x} = \text{vec}(\mathbf{X}^T) \in \mathbb{C}^{NM}$ and $\mathbf{S}(T) \in \mathbb{C}^{NM \times NM}$ denotes the associated two-level Toeplitz matrix. This block-PSD constraint has dimension $(NM+1) \times (NM+1)$. According to the standard complexity model for interior-point methods (IPM) in generic SDP solvers, the arithmetic complexity of an SDP with d free variables and J linear matrix inequalities (LMIs) of sizes $k_j \times k_j$ is given by: $O\left(\left(1 + \sum_{j=1}^J k_j\right)^{\frac{1}{2}} d \left(d^2 + d \sum_{j=1}^J k_j^2 + \sum_{j=1}^J k_j^3\right)\right)$ ignoring logarithmic factors related to the target accuracy [21]. For DANM approach, where $J = 1$, $k_1 = N + M$, and $d = O(NM)$, the complexity is $O(\max\{N^{4.5}M^2, N^2M^{4.5}\})$. In contrast, for the vectorization-based 2D ANM SDP, where $J = 1$, $k_1 = NM + 1$, and $d = O(NM)$, the complexity increases to $O((NM)^{4.5})$.

III. PROPOSED METHOD AND EXACT RECOVERY

A. Proposed Method

In the multichannel scenario, DANM can also be applied to spectral-sparse signal recovery. For each data matrix $\mathbf{X}_\Omega(l)$, we can obtain frequencies $\{\check{\mathbf{f}}(l), \check{\boldsymbol{\phi}}(l)\}$ through

$$\begin{aligned} \min_{\mathbf{t}_x, \mathbf{t}_y, \tilde{\mathbf{X}}} &\frac{1}{2N} \text{tr}(\mathcal{T} \mathbf{t}_x) + \frac{1}{2M} \text{tr}(\mathcal{T} \mathbf{t}_y) \\ \text{subject to } &\begin{bmatrix} \mathcal{T} \mathbf{t}_y & \tilde{\mathbf{X}}^H \\ \tilde{\mathbf{X}} & \mathcal{T} \mathbf{t}_x \end{bmatrix} \succeq 0, \tilde{\mathbf{X}}_\Omega = \mathbf{X}_\Omega(l). \end{aligned} \quad (4)$$

That is to say, we need to solve L optimization problems and select the estimates of true frequencies from $\{\check{\mathbf{f}}(l), \check{\boldsymbol{\phi}}(l)\}_{l=1}^L$. In this way, obviously, DANM cannot utilize the multichannel information that L observation matrices share the same frequencies. Moreover, the CM property is ignored.

To utilize the CM property, we propose the following optimization problem for multichannel signal recovery:

$$\begin{aligned} \min_{\mathbf{t}_x, \mathbf{t}_y, \{\tilde{\mathbf{X}}(l)\}} &\frac{1}{2N} \text{tr}(\mathcal{T} \mathbf{t}_x) + \frac{1}{2M} \text{tr}(\mathcal{T} \mathbf{t}_y) \\ \text{subject to } &\begin{bmatrix} \mathcal{T} \mathbf{t}_y & \tilde{\mathbf{X}}^H(l) \\ \tilde{\mathbf{X}}(l) & \mathcal{T} \mathbf{t}_x \end{bmatrix} \succeq 0, \\ &\tilde{\mathbf{X}}_\Omega(l) = \mathbf{X}_\Omega(l), l \in [L]. \end{aligned} \quad (5)$$

The new algorithm is proposed based on DANM with some significant modifications. All observation matrices are embedded in L SDP constraints that share the same Toeplitz matrices $\mathcal{T}\mathbf{t}_x$ and $\mathcal{T}\mathbf{t}_y$. According to the column inclusion property of PSD matrices, we can obtain that $\tilde{\mathbf{X}}(l) \in \text{range}(\mathcal{T}\mathbf{t}_x)$ and $\tilde{\mathbf{X}}(l)^H \in \text{range}(\mathcal{T}\mathbf{t}_y)$. It implies that the shared frequencies are contained in $\mathcal{T}\mathbf{t}_x$ and $\mathcal{T}\mathbf{t}_y$.

Furthermore, we argue that the CM structure is favored by (5). The following CM-favoring interpretation is provided for the rank-deficient regime, which is the relevant setting under the sparse line spectrum model. Let $\{\mathbf{t}_x, \mathbf{t}_y\}$ be a feasible solution of (5). Assume that the associated Toeplitz matrices $\mathcal{T}\mathbf{t}_x$ and $\mathcal{T}\mathbf{t}_y$ are rank-deficient. By the Vandermonde decomposition of PSD Toeplitz matrices, there exist frequency sets $\hat{\mathbf{f}}$, $\hat{\phi}$ and diagonal matrices $\tilde{\mathbf{P}}_y \succ 0$, $\tilde{\mathbf{P}}_x \succ 0$, such that $\mathcal{T}\mathbf{t}_x = \mathbf{A}_N(\hat{\mathbf{f}})\tilde{\mathbf{P}}_x\mathbf{A}_N(\hat{\mathbf{f}})^H$ and $\mathcal{T}\mathbf{t}_y = \mathbf{A}_M(\hat{\phi})\tilde{\mathbf{P}}_y\mathbf{A}_M(\hat{\phi})^H$, where $r_x = \text{rank}(\mathcal{T}\mathbf{t}_x)$ and $r_y = \text{rank}(\mathcal{T}\mathbf{t}_y)$. Moreover, the above decompositions are unique up to permutation. For notational convenience in the following argument, we rewrite both decompositions using a common index set $k = 1, \dots, r$, where $r = \max\{r_x, r_y\}$. This is achieved by padding the smaller decomposition with zero-weight atoms, which does not change $\mathcal{T}\mathbf{t}_x$ or $\mathcal{T}\mathbf{t}_y$. When performing this padding, we choose the additional (zero-weight) frequencies to be distinct from those already present, so that the resulting Vandermonde factors are full column rank (with $r \leq \min(N, M)$). Under this convention, we write $\mathbf{P}_x = \text{diag}(\mathbf{p}_x) \succeq 0$ and $\mathbf{P}_y = \text{diag}(\mathbf{p}_y) \succeq 0$ and let $\mathbf{A}_N(\hat{\mathbf{f}}) \in \mathbb{C}^{N \times r}$, $\mathbf{A}_M(\hat{\phi}) \in \mathbb{C}^{M \times r}$ denote the corresponding Vandermonde factors. Their Moore–Penrose pseudo-inverses therefore satisfy $\mathbf{A}_N(\hat{\mathbf{f}})^\dagger \mathbf{A}_N(\hat{\mathbf{f}}) = \mathbf{I}_r$ and $\mathbf{A}_M(\hat{\phi})^\dagger \mathbf{A}_M(\hat{\phi}) = \mathbf{I}_r$. Moreover, the objective can be written as:

$$\begin{aligned} \frac{1}{2N}\text{tr}(\mathcal{T}\mathbf{t}_x) + \frac{1}{2M}\text{tr}(\mathcal{T}\mathbf{t}_y) &= \sum_k^r \frac{(\mathbf{p}_y)_k + (\mathbf{p}_x)_k}{2} \\ &\geq \sum_k^r \sqrt{(\mathbf{p}_y)_k(\mathbf{p}_x)_k}. \end{aligned}$$

Pre- and post-multiplying the PSD constraint in (5) by $\mathbf{B} = \begin{bmatrix} \mathbf{A}_M(\hat{\phi})^\dagger & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_N(\hat{\mathbf{f}})^\dagger \end{bmatrix}$, yields

$$\mathbf{B} \begin{bmatrix} \mathcal{T}\mathbf{t}_y & \tilde{\mathbf{X}}^H(l) \\ \tilde{\mathbf{X}}(l) & \mathcal{T}\mathbf{t}_x \end{bmatrix} \mathbf{B}^H = \begin{bmatrix} \mathbf{P}_y & \mathbf{C}(l)^H \\ \mathbf{C}(l) & \mathbf{P}_x \end{bmatrix} \succeq 0,$$

where $\mathbf{C}(l) = \mathbf{A}_N(\hat{\mathbf{f}})^\dagger \tilde{\mathbf{X}}(l)(\mathbf{A}_M(\hat{\phi})^\dagger)^H$. Considering the 2×2 principal submatrix indexed by k gives $\begin{bmatrix} (\mathbf{p}_y)_k & C_{kk}(l) \\ C_{kk}(l) & (\mathbf{p}_x)_k \end{bmatrix} \succeq 0$, which is equivalent to $|C_{kk}(l)| \leq \sqrt{(\mathbf{p}_y)_k(\mathbf{p}_x)_k}$. Define $w_k = \sqrt{(\mathbf{p}_y)_k(\mathbf{p}_x)_k}$. Since $\mathbf{P}_x$ and $\mathbf{P}_y$ are shared across all channels, w_k acts as a common magnitude budget and must satisfy $w_k \geq |C_{kk}(l)|$ for all l . Consequently, the objective is lower bounded by

$$\frac{1}{2N}\text{tr}(\mathcal{T}\mathbf{t}_x) + \frac{1}{2M}\text{tr}(\mathcal{T}\mathbf{t}_y) \geq \sum_k^r w_k \geq \sum_k^r \max_{l \in [L]} |C_{kk}(l)|.$$

This shows that (5) penalizes the largest (across-channel) magnitude associated with each recovered component, and therefore favors CM-like solutions in which the magnitudes are more balanced across channels. Moreover, in the noiseless case where (5) achieves exact recovery $\tilde{\mathbf{X}}(l) = \mathbf{X}(l)$ and the recovered frequencies are correct (with aligned indexing), we have $\mathbf{C}(l) = \mathbf{A}_N(\hat{\mathbf{f}})^\dagger \tilde{\mathbf{X}}(l)(\mathbf{A}_M(\hat{\phi})^\dagger)^H = \mathbf{A}_N(\hat{\mathbf{f}})^\dagger \mathbf{X}(l)(\mathbf{A}_M(\hat{\phi})^\dagger)^H = \text{diag}(\mathbf{s}(l))$. Hence the bound reduces to $\frac{1}{2N}\text{tr}(\mathcal{T}\mathbf{t}_x) + \frac{1}{2M}\text{tr}(\mathcal{T}\mathbf{t}_y) \geq \sum_k \max_{l \in [L]} |s_k(l)|$, which is minimized when $|s_k(l)|$ varies little across l ; in particular, the CM condition $|s_k(l)| = b_k$ for all l is most favored.

Furthermore, it is worth noting that when $\mathbf{X}(l)$ are identical for all l , DANM-CM reduces to DANM, which corresponds to the single channel setting.

Under the standard IPM complexity model for generic SDP solvers, the SDP-solving cost of DANM-CM is on the order of $O(\max\{L^{3.5}N^{4.5}M^2, L^{3.5}N^2M^{4.5}\})$, where each of the L LMIs has dimension $(N+M) \times (N+M)$, and the number of free variables is $d = O(LNM)$.

B. Theoretical Guarantee

We next provide a sufficient condition for the DANM-CM algorithm that characterizes its recovery performance. Before that, we denote two finite subsets $\mathcal{H} := \{f_1, \dots, f_K\}$ and $\mathcal{G} := \{\phi_1, \dots, \phi_K\}$. The minimum separation of a finite subset $\mathcal{H} \subset [0, 1)^K$ as the closest wrap-around distance between any two components is defined as: $\Delta_{\mathcal{H}} = \inf_{a, b \in \mathcal{H}: a \neq b} \min\{|a - b|, 1 - |a - b|\}$. The following proposition is a corollary to [20, Theorem 4.2].

Proposition 1: Suppose that $\mathbf{X}(l) = \mathbf{A}_N(\mathbf{f})\text{diag}(\mathbf{s}(l))\mathbf{A}_M(\phi)^T \in \mathbb{C}^{N \times M}$, $|s_k(l)| = b_k$, $k \in [K]$, $l \in [L]$. If $K \leq \min\{N, M\} - 1$ and these frequencies $\{\mathbf{f}, \phi\}$ satisfy $\Delta_{\mathcal{H}} \geq \frac{1.19}{[(N-1)/4]}$ or $\Delta_{\mathcal{G}} \geq \frac{1.19}{[(M-1)/4]}$, then $\{\mathbf{f}, \phi\}$ can be uniquely recovered from the optimal solution to the following optimization problem

$$\begin{aligned} \min_{\mathbf{t}_x, \mathbf{t}_y} \quad & \frac{1}{2N}\text{tr}(\mathcal{T}\mathbf{t}_x) + \frac{1}{2M}\text{tr}(\mathcal{T}\mathbf{t}_y) \\ \text{subject to} \quad & \begin{bmatrix} \mathcal{T}\mathbf{t}_y & \mathbf{X}^H(l) \\ \mathbf{X}(l) & \mathcal{T}\mathbf{t}_x \end{bmatrix} \succeq 0, \quad l \in [L]. \end{aligned} \quad (6)$$

Proof: Fix any $l \in [L]$. Consider the DANM problem:

$$\begin{aligned} \min_{\mathbf{t}_x, \mathbf{t}_y} \quad & \frac{1}{2N}\text{tr}(\mathcal{T}\mathbf{t}_x) + \frac{1}{2M}\text{tr}(\mathcal{T}\mathbf{t}_y) \\ \text{subject to} \quad & \begin{bmatrix} \mathcal{T}\mathbf{t}_y & \mathbf{X}^H(l) \\ \mathbf{X}(l) & \mathcal{T}\mathbf{t}_x \end{bmatrix} \succeq 0. \end{aligned} \quad (7)$$

Under the separation condition in Proposition 1, it follows from [20, Theorem 4.2] that the true frequencies $\{\mathbf{f}, \phi\}$ can be uniquely recovered from the optimal solution of (7), and moreover the optimizer is unique. Denote this unique optimizer by $(\mathbf{t}_x^0, \mathbf{t}_y^0)$, for which the associated Toeplitz matrices admit the Vandermonde forms

$$\begin{aligned} \mathcal{T}\mathbf{t}_x^0 &= \mathbf{A}_N(\mathbf{f})\text{diag}(|\mathbf{s}(l)|)\mathbf{A}_N(\mathbf{f})^H, \\ \mathcal{T}\mathbf{t}_y^0 &= \mathbf{A}_M(-\phi)\text{diag}(|\mathbf{s}(l)|)\mathbf{A}_M(-\phi)^H. \end{aligned}$$

By the CM assumption, $|s_k(l)| = b_k$ for all $k \in [K]$ and $l \in [L]$, the matrices $\mathcal{T}\mathbf{t}_x^0$ and $\mathcal{T}\mathbf{t}_y^0$ are independent of l . Hence the pair $(\mathbf{t}_x^0, \mathbf{t}_y^0)$ is the unique optimizer of (7) for all $l \in [L]$. Define $g(\mathbf{t}_x, \mathbf{t}_y) = \frac{1}{2N}\text{tr}(\mathcal{T}\mathbf{t}_x) + \frac{1}{2M}\text{tr}(\mathcal{T}\mathbf{t}_y)$ and let $(\tilde{\mathbf{t}}_x, \tilde{\mathbf{t}}_y)$ be any feasible solution of (6).

We next show that $(\mathbf{t}_x^0, \mathbf{t}_y^0)$ is also the unique optimal solution of (6). Specifically, we prove that $(\mathbf{t}_x^0, \mathbf{t}_y^0)$ is a feasible solution of (6) and $g(\mathbf{t}_x^0, \mathbf{t}_y^0) \leq g(\tilde{\mathbf{t}}_x, \tilde{\mathbf{t}}_y)$.

We first prove $(\mathbf{t}_x^0, \mathbf{t}_y^0)$ is also a feasible solution of (6). For each l , it holds that

$$\begin{bmatrix} \mathcal{T}\mathbf{t}_y^0 & \mathbf{X}^H(l) \\ \mathbf{X}(l) & \mathcal{T}\mathbf{t}_x^0 \end{bmatrix} = \mathbf{B} \begin{bmatrix} \text{diag}(|s(l)|) & \text{diag}(\bar{s}(l)) \\ \text{diag}(s(l)) & \text{diag}(|s(l)|) \end{bmatrix} \mathbf{B}^H \\ = \mathbf{B}\mathbf{F}^{-1} \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \text{diag}(|s(l)|) \end{bmatrix} (\mathbf{F}^{-1})^H \mathbf{B}^H \succeq \mathbf{0},$$

where $\mathbf{B} = \begin{bmatrix} \mathbf{A}_M(-\phi) & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_N(f) \end{bmatrix}$ and $\mathbf{F} = \begin{bmatrix} \mathbf{I} & -\text{diag}(\bar{s}(l))(\text{diag}(|s(l)|))^{-1} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$. Hence, $(\mathbf{t}_x^0, \mathbf{t}_y^0)$ is a feasible solution for (6).

We next prove $g(\mathbf{t}_x^0, \mathbf{t}_y^0) \leq g(\tilde{\mathbf{t}}_x, \tilde{\mathbf{t}}_y)$. Compared with the SDP in (7), problem (6) imposes additional constraints. Hence, the feasible set of (6) is contained in that of (7). Moreover, $(\mathbf{t}_x^0, \mathbf{t}_y^0)$ is the optimal solution of (7), thus $g(\mathbf{t}_x^0, \mathbf{t}_y^0) \leq g(\tilde{\mathbf{t}}_x, \tilde{\mathbf{t}}_y)$.

Finally, we prove uniqueness. Assume that $(\hat{\mathbf{t}}_x, \hat{\mathbf{t}}_y) \neq (\mathbf{t}_x^0, \mathbf{t}_y^0)$ is another optimal solution of (6), i.e. $g(\hat{\mathbf{t}}_x, \hat{\mathbf{t}}_y) = g(\mathbf{t}_x^0, \mathbf{t}_y^0)$. Then, $(\hat{\mathbf{t}}_x, \hat{\mathbf{t}}_y)$ is feasible for (7) and achieves the same objective value as $(\mathbf{t}_x^0, \mathbf{t}_y^0)$. This contradicts the fact that $(\mathbf{t}_x^0, \mathbf{t}_y^0)$ is the unique optimizer of (7). $\square$

The proof of Proposition 1 shows that when the separation condition is satisfied, DANM can exactly recover the true frequencies, and it follows that DANM-CM also succeeds in frequency recovery.

IV. NUMERICAL SIMULATION

In this section, numerical results are presented to illustrate the effectiveness of DANM-CM in 2D LSE with the CM property. We use DANM for comparison. As convex optimization problems, DANM and DANM-CM are implemented in CVX [22] and solved by the SDPT3 solver [23]. The root mean square error (RMSE) is defined as $\sqrt{\frac{\sum_{k=1}^K (|f_k - \hat{f}_k|^2 + |\phi_k - \hat{\phi}_k|^2)}{K}}$. We say that the frequencies are successfully recovered if $\text{RMSE} < 10^{-4}$. Theoretically, the matrices $\mathcal{T}\mathbf{t}_x, \mathcal{T}\mathbf{t}_y$ containing the true frequencies and amplitudes have rank K . With L channels, DANM yields L matrix pairs $\{\mathcal{T}\mathbf{t}_x(l), \mathcal{T}\mathbf{t}_y(l)\}$. In the simulations, we select as the final output the pair whose effective rank is closest to K , and use it for subsequent frequency recovery. The observed matrix $\mathbf{X}_\Omega(l)$ is obtained from the full data matrix $\mathbf{X}(l) \in \mathbb{C}^{N \times M}$ by randomly sampling MNp entries (rounded up to the nearest integer) out of the total MN , where p denotes the sampling rate. The true frequencies are randomly generated with a minimum separation of Δf . The signal amplitudes are modeled as

random complex values with magnitudes $1 + |\mathcal{N}(0, 1)|$ and phases uniformly distributed over $U[0, 2\pi)$. We perform 100 Monte Carlo trials in each experiment.

In *Experiment 1*, we examine the behavior of DANM-CM with varying K and L . The parameters are set to $M = N = 10$, $p = 0.6$, and $\Delta f = 0.5/N$. As illustrated in Fig. 1, DANM-CM achieves a considerably larger recovery region than DANM, demonstrating the effective utilization of the CM property.

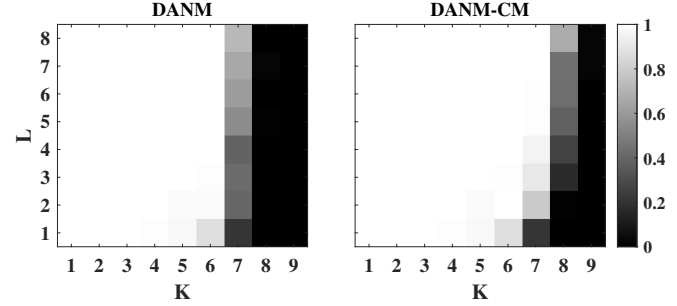


Fig. 1. The phase transition result of DANM and DANM-CM, when $M = N = 10$, $p = 0.6$ and $\Delta f = \frac{0.5}{N}$.

In *Experiment 2*, we evaluate the ability of DANM-CM to exploit the CM structure of signals with respect to p and Δf . The parameters are set to $K = 5$ and $L = 5$, while the other settings remain the same as in Experiment 1. As shown in Fig. 2, DANM-CM consistently outperforms DANM when the frequency separation is small, with the performance gap being particularly evident at $p = 0.4$.

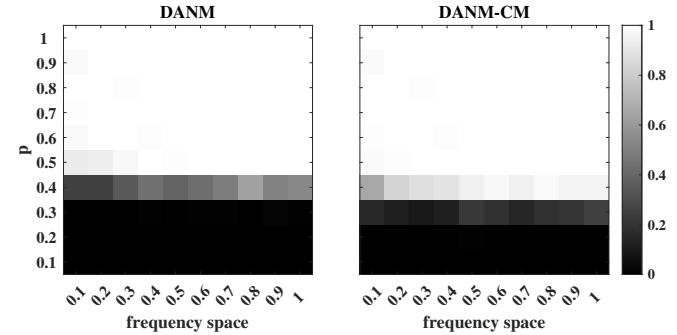


Fig. 2. The phase transition result of DANM and DANM-CM, when $K = 5$ and $L = 5$. The horizontal axis reports the normalized frequency separation $N\Delta f$.

V. CONCLUSION

In this paper, the DANM-CM was proposed for 2D line spectral estimation with the CM property. We proved that, under a frequency separation condition, DANM-CM guarantees exact recovery of the true frequencies. Extensive numerical simulations were conducted to showcase the effectiveness of the proposed method.

REFERENCES

- [1] R. Heckel and M. Soltanolkotabi, "Generalized line spectral estimation via convex optimization," *IEEE Transactions on Information Theory*, vol. 64, no. 6, pp. 4001–4023, 2018.
- [2] Z. Gao, L. Dai, Z. Lu, C. Yuen, and Z. Wang, "Super-resolution sparse MIMO-OFDM channel estimation based on spatial and temporal correlations," *IEEE Communications Letters*, vol. 18, no. 7, pp. 1266–1269, 2014.
- [3] B. Huang, W. Wang, M. Bates, and X. Zhuang, "Three-dimensional super-resolution imaging by stochastic optical reconstruction microscopy," *Science*, vol. 319, no. 5864, pp. 810–813, 2008.
- [4] W. Sharif, M. Muma, and A. M. Zoubir, "Robustness analysis of spatial time-frequency distributions based on the influence function," *IEEE Transactions on Signal Processing*, vol. 61, no. 8, pp. 1958–1971, 2013.
- [5] Z. Hamici, "Elements failure robust compensation in 2D phased arrays for DOA estimation with M-ary PSK signals," *IEEE Transactions on Wireless Communications*, vol. 19, no. 2, pp. 1347–1356, 2020.
- [6] A. Leshem and A.-J. van der Veen, "Direction-of-arrival estimation for constant modulus signals," *IEEE Transactions on Signal Processing*, vol. 47, no. 11, pp. 3125–3129, 1999.
- [7] M. Wax, "On unique localization of constrained-signals sources," *IEEE Transactions on Signal Processing*, vol. 40, no. 6, pp. 1542–1547, 1992.
- [8] S. Valaee and P. Kabal, "Alternative proofs for "On unique localization of constrained-signal sources"," *IEEE Transactions on Signal Processing*, vol. 42, no. 12, pp. 3547–3549, 1994.
- [9] X. Wu, Z. Yang, Z. Wei, and Z. Xu, "Direction-of-arrival estimation for constant modulus signals using a structured matrix recovery technique," *IEEE Transactions on Wireless Communications*, pp. 1–1, 2023.
- [10] X. Wu, Z. Yang, and Z. Xu, "Multichannel frequency estimation with constant amplitude via convex structured low-rank approximation," *SIAM Journal on Matrix Analysis and Applications*, vol. 45, no. 3, pp. 1643–1668, 2024.
- [11] Y.-L. Mo, W. Wang, J. Shi, and Z. Yang, "Multichannel sparse recovery for constant modulus signals via $\ell_{\infty,1}$ minimization," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 4, pp. 9761–9773, 2025.
- [12] A.-J. van der Veen and A. Paulraj, "An analytical constant modulus algorithm," *IEEE Transactions on Signal Processing*, vol. 44, no. 5, pp. 1136–1155, 1996.
- [13] Y. Sato, "A method of self-recovering equalization for multilevel amplitude-modulation systems," *IEEE Transactions on Communications*, vol. 23, no. 6, pp. 679–682, 1975.
- [14] J. Treichler and B. Agee, "A new approach to multipath correction of constant modulus signals," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 31, no. 2, pp. 459–472, 1983.
- [15] Y. Hua, "A pencil-music algorithm for finding two-dimensional angles and polarizations using crossed dipoles," *IEEE Transactions on Antennas and Propagation*, vol. 41, no. 3, pp. 370–376, 1993.
- [16] M. Haardt, M. Zoltowski, C. Mathews, and J. Nosssek, "2D unitary esprit for efficient 2D parameter estimation," *1995 International Conference on Acoustics, Speech, and Signal Processing*, vol. 3, 1995, pp. 2096–2099 vol.3.
- [17] Y. Fang, J. Wu, and B. Huang, "2D sparse signal recovery via 2D orthogonal matching pursuit," *Science China Information Sciences*, vol. 55, no. 4, pp. 889–897, 2012.
- [18] C. Zhang, Q. Chen, M. Wang, and S. Wei, "Optimised two-dimensional orthogonal matching pursuit algorithm via singular value decomposition," *IET Signal Processing*, vol. 14, no. 10, pp. 717–724, 2020.
- [19] Y. Chi and Y. Chen, "Compressive two-dimensional harmonic retrieval via atomic norm minimization," *IEEE Transactions on Signal Processing*, vol. 63, no. 4, pp. 1030–1042, 2015.
- [20] Z. Zhang, Y. Wang, and Z. Tian, "Efficient two-dimensional line spectrum estimation based on decoupled atomic norm minimization," *Signal Processing*, vol. 163, pp. 95–106, 2019.
- [21] A. Ben-Tal and A. Nemirovski, *Lectures on modern convex optimization: analysis, algorithms, and engineering applications*. SIAM, 2001.
- [22] M. Grant, S. Boyd, and Y. Ye, "CVX: Matlab software for disciplined convex programming," 2008.
- [23] R. H. Tütüncü, K.-C. Toh, and M. J. Todd, "Sdpt3—a matlab software package for semidefinite-quadratic-linear programming, version 3.0," Web page <http://www.math.nus.edu.sg/mattohkc/sdpt3.html>, 2001.